

II BE Mid Semester Test II 2023
(IT A&B) Engineering
(3ITRC4): (DIGITAL ELECTRONICS)

Duration: 70 min

Maximum Marks : 20

All questions are of equal marks. Attempt any four.

Q.1) Implement a full Subtractor using two half Subtractors and basic gates.

Q.2) Simplify the following function using tabulation method.

$$F(A,B,C,D) = \sum(0,1,6, 7,8,9,13,14,15)$$

Q.3) Implement full adder with basic gates and also draw the logic diagram of full adder using only NAND gates.

Q.4) Design a 4 bit binary to gray code converter.

Q.5) Find the minimal expressions for the function

$$F(A,B,C,D) = \pi(3,4,5,7,11,13,15) + d(6,8,10,12)$$

